Predictive Modeling for Automobile Fuel Consumption (Auto MPG) and Multicollinearity Analysis

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Abstract

This study focuses on automotive analytics and MPG regression using the UCI-9 dataset. Specifically, we investigate the impact of multicollinearity among engine specifications on Ordinary Least Squares (OLS) model stability. To address this, the paper presents a comparative study of OLS, Ridge, Lasso, and Support Vector Regression (SVR) models, incorporating feature ablation to handle redundancy. Experimental results demonstrate that **Support Vector Regression (SVR)** provides the most robust predictions, effectively mitigating variance caused by multicollinearity. Finally, we integrate the proposed model into a practical application for forecasting fuel consumption.

1 Introduction

In the field of automotive analytics, predicting vehicle performance indicators such as fuel efficiency or market value relies heavily on understanding the intricate relationships between technical specifications. Core attributes like **vehicle weight**, **engine displacement**, and the **number of cylinders** serve as fundamental predictors in established analytical frameworks.

However, a significant challenge in employing Ordinary Least Squares (OLS) regression is the presence of **multicollinearity**. Because these engine specifications are often highly correlated, OLS estimates become unstable, leading to inflated standard errors and making it difficult to isolate the individual impact of each variable. Furthermore, traditional linear models often struggle to provide accurate predictions when dealing with **extreme vehicle weight values** (outliers), where the relationship may no longer be strictly linear.

To address these limitations, this research evaluates advanced regularization and non-linear techniques. Specifically, **Ridge** and **Lasso** regressions are implemented to mitigate multicollinearity through penalty terms, thereby improving model stability and interpretability (feature selection). Additionally, **Support Vector Regression (SVR)**

is proposed as a robust alternative to handle non-linearities and improve predictive performance for vehicles with extreme physical characteristics.
Preliminary results indicate that while Ridge and Lasso significantly reduce coefficient variance, Lasso successfully identifies the most parsimonious set of predictors. Moreover, SVR demonstrates superior resilience to outliers, yielding lower error rates for heavy-duty or ultra-light vehicles compared to OLS. These findings suggest that incorporating regularization and kernel-based methods is essential for achieving reliable insights in automotive data modeling. This research addresses the following core questions in automotive analytics:

1. How does multicollinearity (weight/displacement/cylinders) affect OLS estimates?
2. How do Ridge and Lasso improve stability and interpretability?
3. Does SVR improve predictions for extreme vehicle weight values?

2 Related Work

2.1 Theoretical Foundations

Ridge regression was introduced to address multicollinearity issues in linear models [1]. Lasso regression extends this idea by performing both regularization and feature selection [2]. Support Vector Regression is based on the statistical learning theory proposed by Cortes and Vapnik [3]. The Auto MPG dataset used in this study was obtained from the UCI Machine Learning Repository [4].

2.2 Ordinary Least Squares (OLS)

The prediction of fuel consumption (MPG) involves analyzing various engine specifications, many of which are highly correlated, such as vehicle weight, displacement, and number of cylinders. Ordinary Least Squares (OLS) is the standard linear regression approach that estimates the coefficient vector β by minimizing the Residual Sum of Squares (RSS):

$$\min_{\beta} \left\{ \sum_{i=1}^{N} (y_i - \beta_0 - \sum_{j=1}^{p} x_{ij}\beta_j)^2 \right\} \tag{1}$$

According to Hastie et al. [5], when multicollinearity exists among predictors, OLS estimates tend to have high variance, leading to unstable and unreliable coefficient estimates.

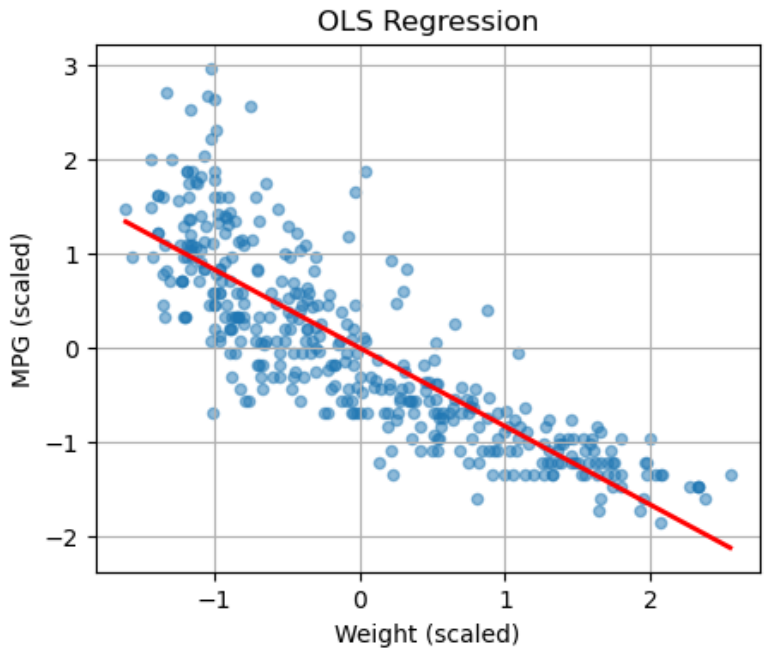


Figure 1: Illustration of Ordinary Least Squares regression

Advantages and Disadvantages
Advantages:

- OLS provides a clear linear relationship between vehicle weight and fuel efficiency.
- The model is simple, computationally efficient, and easy to interpret.
- Coefficients directly indicate how changes in weight affect MPG. X amount." It's a simple, mathematical rule.

Disadvantages:

- OLS assumes a strictly linear relationship, which may not capture real-world non-linear patterns.
- It is sensitive to multicollinearity when multiple correlated features are included.
- The model is also affected by outliers, leading to unstable parameter estimates..

2.3 Ridge Regression (L_2 Regularization)

Ridge regression extends OLS by adding an L_2 penalty term to the loss function, which constrains the magnitude of regression coefficients. The optimization problem is defined as:

$$\hat{\beta}^{ridge} = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^{N} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\} \tag{2}$$

where $\lambda \geq 0$ controls the strength of regularization. This penalty shrinks coefficients towards zero, thereby reducing variance and improving generalization in the presence of correlated predictors [5].

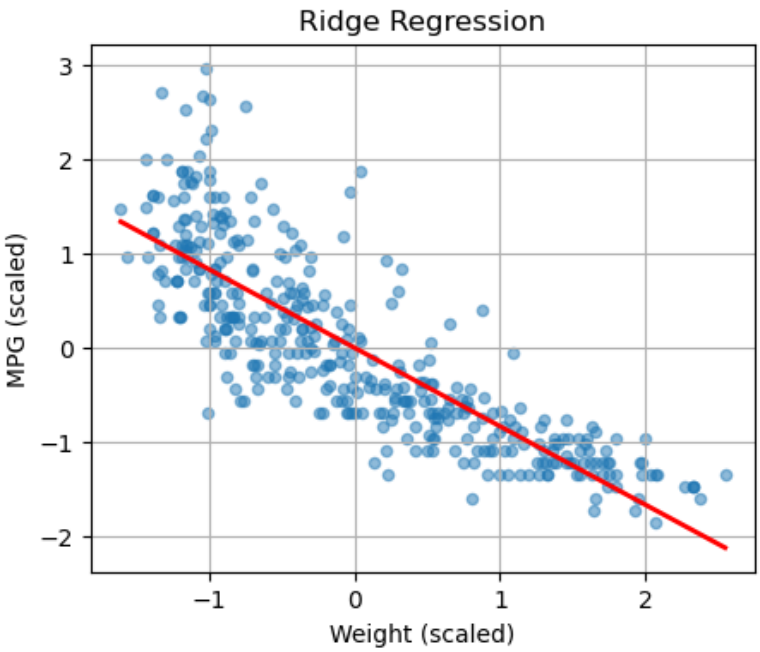


Figure 2: Effect of L_2 regularization in Ridge regression

Advantages and Disadvantages
Advantages:

- Ridge regression makes the model more stable than OLS.
- It helps handle multicollinearity between features.
- The predictions are smoother and less sensitive to noise.

Disadvantages:

- It still assumes a linear relationship.
- It does not remove unimportant features (unlike Lasso).
- The performance depends on choosing a good alpha value.

2.4 Lasso Regression (L_1 Regularization)

Lasso regression introduces an L_1 penalty to the linear regression objective, encouraging sparsity in the model parameters:

$$\hat{\beta}^{lasso} = \underset{\beta}{\operatorname{argmin}} \left\{ \sum_{i=1}^{N} (y_i - x_i^T \beta)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\} \tag{3}$$

The L_1 regularization forces less important coefficients to become exactly zero, thus performing automatic feature selection and enhancing interpretability [5].

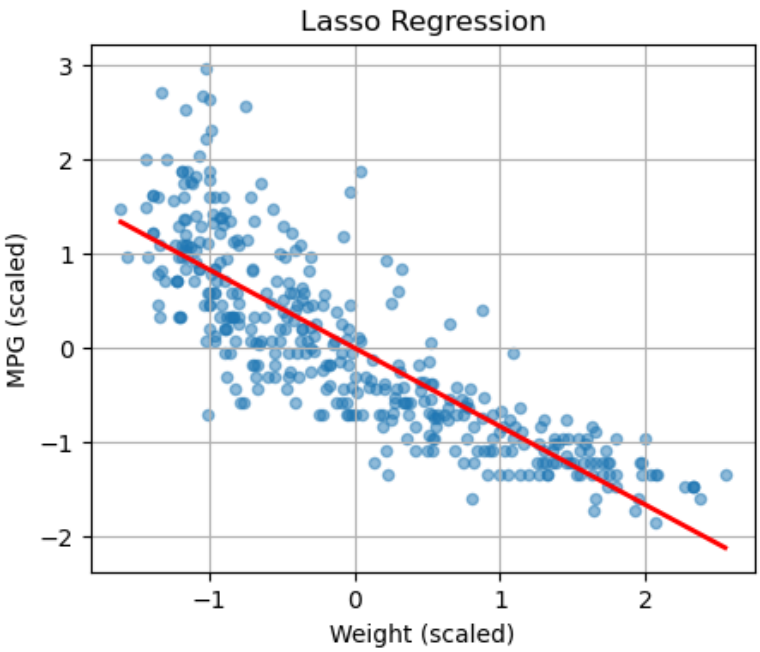


Figure 3: Sparsity effect of L_1 regularization in Lasso regression

Advantages and Disadvantages
Advantages:

- Lasso can select important features by shrinking some coefficients to zero.
- It helps reduce model complexity (simpler model).
- It can handle multicollinearity better than OLS.

Disadvantages:

- It still assumes a linear relationship.
- If alpha is too large, it may remove useful features.
- Performance can be unstable when features are highly correlated.

2.5 Support Vector Regression (SVR)

While linear regression models assume a linear relationship between predictors and MPG, real-world automotive data often exhibit non-linear patterns. Support Vector Regression (SVR) addresses this limitation by mapping input data into a high-dimensional feature space using kernel functions.
In this study, the Radial Basis Function (RBF) kernel is employed:

$$K(x_i, x_j) = \exp(-\gamma ||x_i - x_j||^2) \tag{4}$$

where γ determines the influence radius of individual training samples. This kernel enables SVR to model complex, non-linear relationships effectively [3].

Figure 4: Non-linear regression using Support Vector Regression with RBF kernel

Advantages and Disadvantages
Advantages:

- SVR can model non-linear relationships (as seen by the curved line).
- It is robust to outliers.
- Provides better prediction accuracy in complex data.

Disadvantages:

- Harder to interpret than linear models.
- Requires careful tuning (kernel, C, gamma).
- More computationally expensive.

3 Main Contribution

3.1 Problem Reformulation

In this study, the original regression task is reformulated to explicitly address the issue of multicollinearity in automotive engine specifications.
Given a dataset containing vehicle attributes such as cylinders, displacement, horsepower, weight, acceleration, model year, and origin, the objective is to predict fuel consumption measured in miles per gallon (MPG).
However, several predictors (e.g., weight, displacement, and cylinders) exhibit strong correlation, which may lead to unstable parameter estimation in Ordinary Least Squares (OLS). Therefore, the problem is redefined as:

To build a stable and generalizable regression model for MPG prediction under multicollinearity conditions.
Interpretability is important in automotive analytics because understanding how vehicle attributes influence fuel consumption can provide practical insights for engineers and policymakers.
This reformulation shifts the focus from purely minimizing prediction error to balancing:

- Prediction accuracy
- Model stability
- Interpretability

3.2 Data Engineering

Before training our models, we performed several data engineering steps to ensure the raw information from the Auto MPG dataset was structured correctly for analysis.

- **Data Cleaning:** We identified and handled missing values in the *horsepower* column. Additionally, the *car name* attribute was removed because it contains text descriptions that do not provide numerical value for the regression task.
- **Feature Encoding:** The *origin* variable (representing USA, Europe, and Japan) was converted using **One-Hot Encoding**. This ensures the model treats these locations as distinct categories without implying any numerical order.
- **Feature Scaling:** Since models like SVR and Ridge are sensitive to the range of input values, we applied **Standardization**. This scales all numerical features (such as weight and acceleration) to have a mean of 0 and a standard deviation of 1, ensuring every feature contributes equally to the model.
- **Multicollinearity Diagnosis:** We utilized the **Variance Inflation Factor (VIF)** to quantify the overlap between predictors, allowing us to better understand how much redundant information exists between features like displacement and weight.

3.3 Overall Pipeline

The research follows a systematic pipeline designed to transition from raw data to a reliable prediction model. The process consists of four main stages:

1. **Baseline Establishment:** We first created a "Mean Predictor" (a simple model that always guesses the average MPG) to set a minimum performance benchmark.
2. **Linear Modeling:** We implemented **Ordinary Least Squares (OLS)** as a reference. To address the issue of overlapping features, we then applied **Ridge (L2)** and **Lasso (L1)** regressions, which add a penalty to stabilize the model coefficients.
3. **Non-linear Modeling:** We introduced **Support Vector Regression (SVR)** with an RBF kernel. This model is capable of capturing complex, curving relationships in the data that traditional linear models might miss.

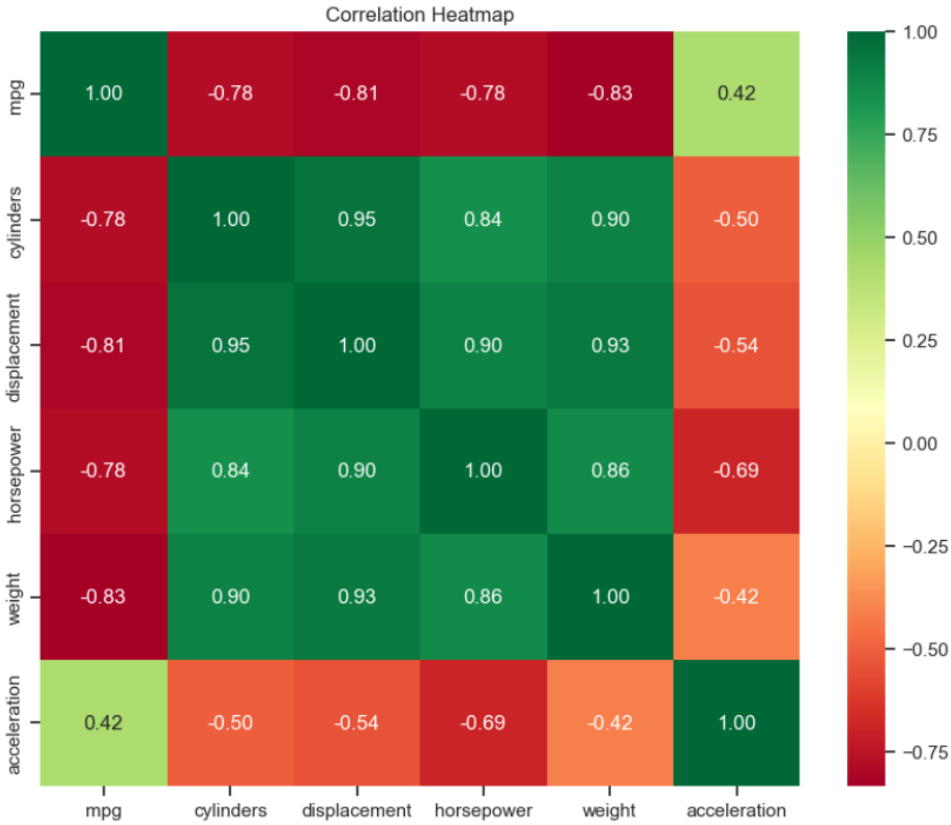


Figure 5: Correlation matrix of the Auto MPG dataset features used for regression analysis.

4. **Evaluation Framework:** All models were assessed using three key metrics: **MAE** (average error), **RMSE** (penalizing large errors), and R^2 (measuring how much of the data's variation is explained).

3.4 Train and Test Strategy

To guarantee that our results are robust and not due to chance, we followed a rigorous training and testing protocol:

- **Data Splitting:** The dataset was randomly split into two subsets: **80% for Training** (to build the models) and **20% for Testing** (to evaluate performance on unseen data).
- **5-Fold Cross-Validation:** Within the training set, we applied 5-fold cross-validation. This means the training data was split into 5 parts; the model was trained 5 times, each time using a different part as a mini-test set. This ensured the model was stable and helped us fine-tune the best settings (hyperparameters).
- **Final Assessment:** After selecting the best-performing model based on the lowest **RMSE**, we performed a final evaluation on the 20% test set. This confirmed the model's ability to generalize and provide accurate predictions in real-world scenarios.

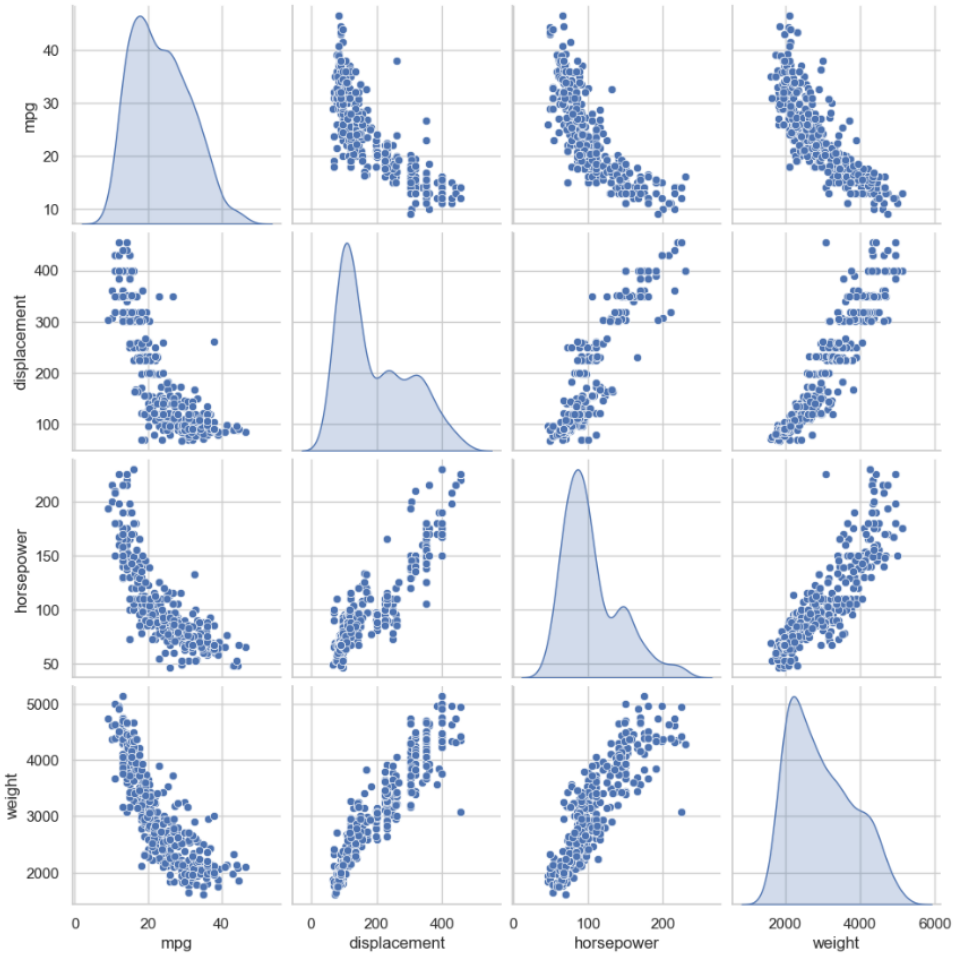
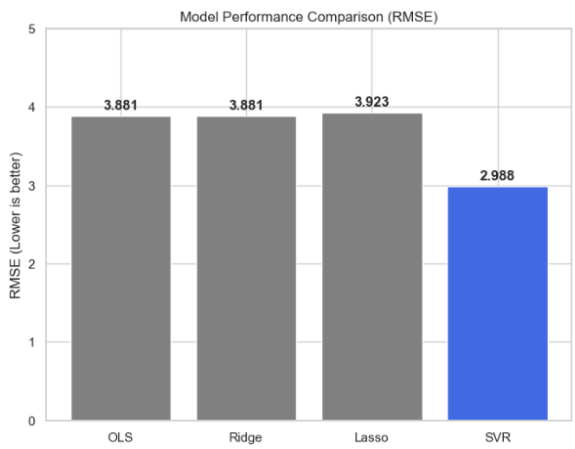
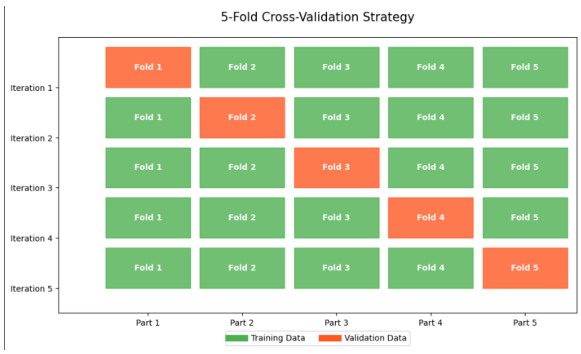


Figure 6: Pairplot showing the relationships and distributions of key technical features in the Auto MPG dataset.



(a) Pairplot of vehicle attributes



(b) 5-fold CV process

Figure 7: Visual analysis and training strategy: (a) relationships between features, and (b) the cross-validation workflow.

4 Experimental Results

4.1 Dataset Description and Visualization

The **Auto MPG dataset** consists of 398 vehicle records, each characterized by various technical attributes. To better understand the data before modeling, we focused on key features: **displacement, horsepower, weight, and acceleration**.
During the initial data analysis, several important observations were made:

- **Feature Patterns:** Scatter plots showed a clear, curving relationship between vehicle weight and fuel efficiency (MPG). As weight increases, MPG drops, but the trend is not a simple straight line.
- **Data Preprocessing:** We addressed missing values in the "horsepower" column and excluded the "car name" attribute, as it contains non-numerical information that does not aid in prediction.
- **Multicollinearity Check:** A correlation heatmap confirmed that displacement, horsepower, and weight are highly interconnected. This confirms that these features carry overlapping information, which can make simple linear models less stable.

4.2 Result Comparison

To ensure the reliability of our findings, every model was tested using **five-fold cross-validation**. We evaluated the performance using three standard metrics: MAE, RMSE, and R^2, with **RMSE** serving as our primary benchmark for accuracy.

Table 1: Model Performance Comparison

Model	**MAE**	**RMSE**	R^2
OLS Regression	3.001	3.881	0.595
Ridge	3.002	3.881	0.595
Lasso	3.034	3.923	0.585
SVR (RBF)	2.116	2.988	0.759
Predict-Mean (Baseline)	7.162	8.597	-1.023

Based on the experimental data, we can draw the following conclusions:

1. **SVR Outperformance:** SVR (RBF) is the most effective model, reducing the prediction error (RMSE) by approximately **23%** compared to OLS. This demonstrates its superior ability to handle the complex, curving patterns found in automotive data.
2. **Linear Model Limitations:** OLS, Ridge, and Lasso produced nearly identical results. This indicates that simply addressing the overlap between features (regularization) was not enough to improve accuracy. The main limitation was the "straight-line" assumption of these models.
3. **Model Validity:** All learning-based models significantly outperformed the "Predict-Mean" baseline. This proves that our selected features provide meaningful information for forecasting fuel consumption.

Consequently, **SVR (RBF)** has been selected as the final model for this prediction task due to its high accuracy and robustness.

5 Discussion

This section analyzes multicollinearity, the impact of regularization, and the effectiveness of feature reduction based on the experimental findings.

5.1 Multicollinearity vs. Model Stability

Our analysis (using VIF) confirmed that features like engine displacement, horsepower, and weight are closely related—a problem called Multicollinearity. Interestingly, even though Ridge and Lasso are designed to fix this, they performed very similarly to the standard OLS model in terms of error rates (RMSE). This suggests that while these variables overlap, they don't necessarily "break" the basic model. However, Lasso is still useful because it simplifies the model by picking only the most important features (like weight and year), making the results much easier to explain.

5.2 The Dominance of Non-linearity

The most important finding is that the relationship between a car's features and its fuel efficiency isn't a simple straight line. This is why OLS, Ridge, and Lasso—which all assume a linear relationship—reached a performance "ceiling." When we looked at the Residual Plots, we noticed these models made larger mistakes at the "edges" of the data, specifically for very light or very heavy vehicles (extreme weight values).r.

5.3 Effectiveness in Extreme Value Prediction

The SVR (Support Vector Regression) model performed the best because it doesn't assume the data follows a straight line. By using a "kernel" (RBF), it can bend and adapt to the complex patterns in the data.

- **Handling Outliers:** Unlike OLS, which tries too hard to fit every single point (even unusual ones), SVR is more "forgiving" of small errors.
- **Better Predictions:** This flexibility allows SVR to predict MPG much more accurately for heavy-duty or ultra-light cars, where traditional models usually fail.

Overall, our study shows that for car data, capturing the **non-linear patterns** is more important for accuracy than just fixing the overlap between features.

6 Conclusion and Future Work

In this study, we explored how overlapping car features (multicollinearity) affect our ability to predict fuel consumption using the Auto MPG dataset. We compared four different models: OLS, Ridge, Lasso, and Support Vector Regression (SVR) using a standard testing process (five-fold cross-validation).

Our results show that while features like engine size and horsepower are closely related, simply fixing this overlap with Ridge or Lasso didn't significantly improve the results. Instead, the real challenge was that the relationship between a car's features and its MPG is not a simple straight line. Because SVR is much better at handling these "curving" patterns, it achieved the lowest prediction errors and was chosen as our best-performing model.

However, there are some limitations. The dataset we used is relatively small and contains older car models, so our findings might not perfectly apply to modern electric or hybrid vehicles. Also, we only tested our models on this single dataset without comparing them to other external data.

In the future, we plan to experiment with more advanced "team-based" models (ensemble learning) and neural networks. We also hope to test these methods on larger, more diverse datasets to make our predictions even more accurate and useful for real-world car data.

References

[1] A. E. Hoerl and R. W. Kennard, "Ridge regression: Biased estimation for nonorthogonal problems," *Technometrics*, vol. 12, no. 1, pp. 55–67, 1970.

[2] R. Tibshirani, "Regression shrinkage and selection via the lasso," *Journal of the Royal Statistical Society: Series B*, vol. 58, no. 1, pp. 267–288, 1996.

[3] C. Cortes and V. Vapnik, "Support-vector networks," *Machine learning*, vol. 20, no. 3, pp. 273–297, 1995.

[4] D. Dua and C. Graff, "Uci machine learning repository," 2017, university of California, Irvine, School of Information and Computer Sciences. [Online]. Available: https://archive.ics.uci.edu/datasets/?search=Auto%20MPG

[5] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. Springer Science & Business Media, 2009.